

The cover features a dark blue background with a large, stylized white world map in the center. The map is composed of small dots. To the left of the map, there are several overlapping geometric shapes in dark blue and gold, creating a modern, abstract design. The title is centered over the map in a bold, black, sans-serif font. Below the title, the subtitle is centered in a smaller, lighter blue font. The overall aesthetic is professional and tech-oriented.

AI-DRIVEN SALES PIPELINE INTELLIGENCE DASHBOARD

Executive Analytics case study

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Executive Summary

AI-Driven Sales Pipeline Intelligence Dashboard

Project Overview

Organizations generate thousands of sales opportunities every year across multiple products, countries, managers, and sales representatives. Although this information exists inside CRM systems, decision-makers often struggle to transform raw transactional data into meaningful business insights.

This project demonstrates how modern Business Intelligence and Artificial Intelligence techniques can transform sales opportunity data into actionable executive insights.

This project demonstrates the design and development of an AI-driven Sales Pipeline Intelligence Dashboard using Microsoft Power BI. The solution transforms CRM opportunity data into actionable business insights through advanced KPI tracking, interactive visualizations, AI-powered analytics, and executive reporting.

Transforming CRM Data into AI-Powered Business Intelligence for Executive Decision-Making.

Instead of creating static reports, this solution enables leaders to:

- Monitor sales performance
- Analyze product profitability
- Evaluate geographic trends
- Understand sales team effectiveness
- Identify revenue drivers using AI
- Support data-driven strategic decisions

Executive Snapshot

Revenue: 10M

Win Rate: 63.15%

Countries: 15

The project follows industry-standard dimensional modeling practices and modern dashboard design principles commonly used by consulting organizations.

Business Objective

Design an AI-powered Executive Sales Intelligence Dashboard that enables leadership teams to monitor sales performance, identify opportunities, and make data-driven decisions through interactive analytics.

Technologies Used	
Technology	Purpose
Power BI Desktop	Dashboard Development
Power Query	Data Cleaning & Transformation
DAX	Business Calculations
SQL Concepts	Data Modeling
Microsoft Bing Maps	Geographic Analytics
AI Visuals	Key Influencers & Decomposition Tree
Excel	Initial Data Processing

Key Executive Insight: GTK Plus Pro emerged as the highest revenue-generating product, while the United States remained the strongest-performing market with an overall win rate of 63.15%.

Business Problem Statement

Background

Organizations often manage thousands of sales opportunities across products, managers, and countries.

Without centralized analytics, leadership struggles to identify:

- Revenue trends
- Top-performing products
- High-value customers
- Regional performance
- Sales bottlenecks
- Win/Loss patterns

This dashboard addresses these challenges through a centralized AI-powered reporting solution.

Traditional CRM reports generally answer questions such as:

- "What happened?"

However, executives require answers to questions like:

- Why did revenue increase?
- Which product drives maximum profitability?
- Which countries require attention?
- Which sales representatives consistently outperform?
- Where should management invest more resources?

Executive Challenge: Leadership required a unified analytics platform to monitor revenue performance, product intelligence, geographic trends, and sales effectiveness in a single view

Without a centralized analytical platform, answering these questions becomes time-consuming and heavily dependent on manual reporting.

Proposed Solution

An AI-powered executive dashboard capable of delivering:



This enables faster decision-making and improved strategic planning.

Data Model Architecture

Data Model Overview

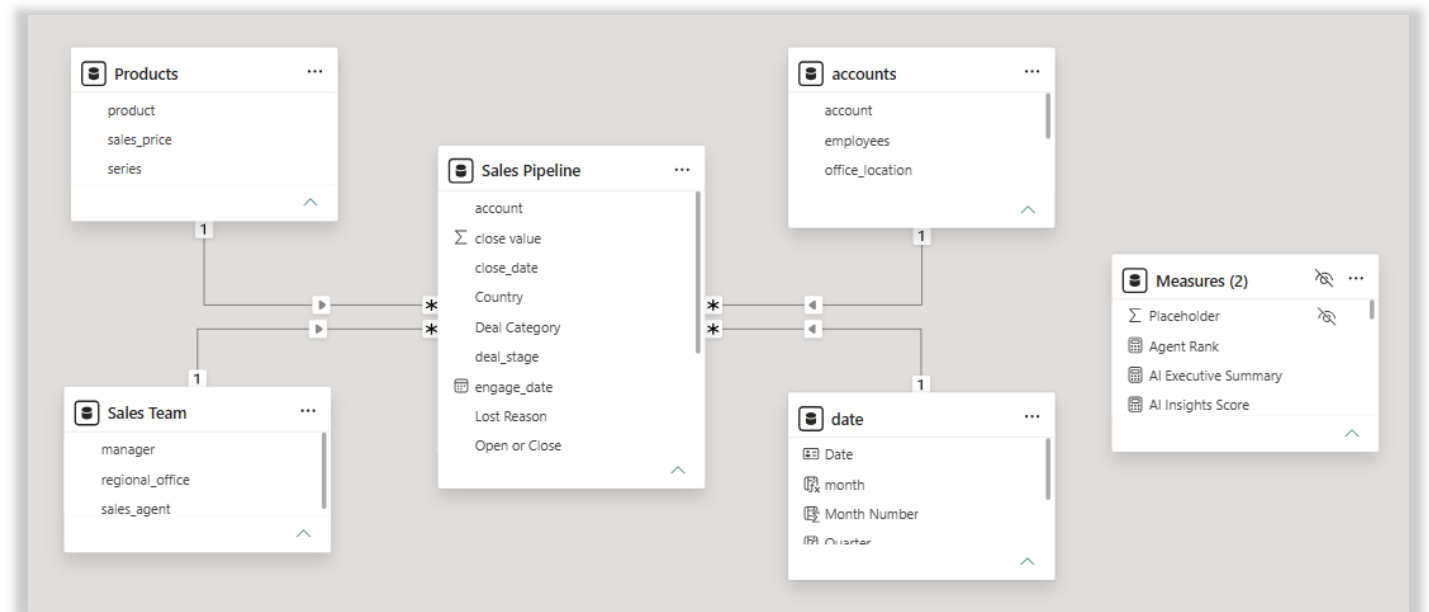


Figure 1: Enterprise Star Schema used for scalable and high-performance Power BI reporting

Star Schema Design

The solution follows a Star Schema data model to maximize query performance and simplify reporting. The Sales Pipeline table acts as the central fact table and is connected to dimension tables including Product, Sales Team, Accounts, and Date.

This architecture minimizes redundancy, improves filter propagation, and aligns with enterprise Power BI best practices.

Fact Table

Sales Pipeline: Contains transactional sales opportunity information including.

Dimension Tables

Products: Provides product hierarchy and pricing information.

Sales Team: Stores sales representatives, managers, and regional office details.

Accounts: Contains customer information including office location and employee count.

Date: Supports time intelligence calculations such as monthly and yearly trends.

Measures: Contains reusable DAX calculations including KPIs and AI measures.

Benefits of this Architecture

- Faster dashboard performance
- Simplified DAX calculations
- Better scalability
- Industry-standard data modelling

Dashboard 01

Executive Overview



Figure 2: Executive Overview Dashboard

Purpose

Provide executives with a consolidated overview of overall sales performance

Key Performance Indicators

- Won Revenue
- Won Deals
- Lost Deals
- Win Rate
- Average Deal Size

Business Insights

The Executive Overview enables leadership to immediately identify overall sales performance without navigating multiple reports.

The monthly trend highlights seasonal fluctuations while product revenue rankings identify top-performing offerings.

The sales conversion funnel visualizes opportunity progression across pipeline stages, helping management monitor conversion effectiveness.

Business Value

This dashboard reduces executive reporting time by consolidating multiple operational reports into one interactive view.

Dashboard 02

Sales Performance Dashboard

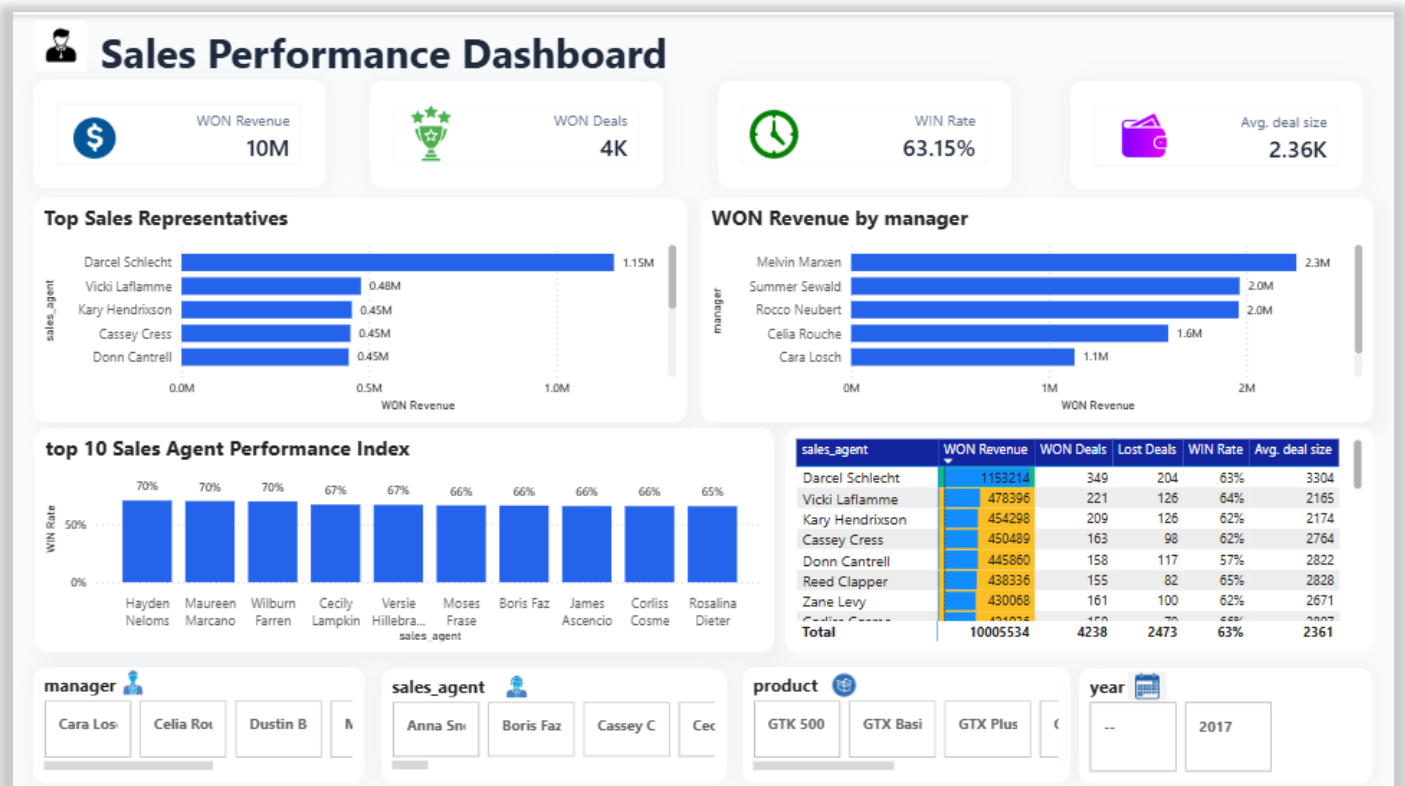


Figure 3: Sales Performance Dashboard

Purpose

Evaluate individual sales representative and manager performance.

Analysis Performed

- Top Performing Sales Representatives
- Manager-wise Revenue
- Sales Agent Performance Index
- Detailed Performance Scorecard
- Interactive Filtering

Key Business Findings

Revenue distribution among representatives shows clear performance variations.

Management can identify:

- Top performers
- Average performers
- Underperforming representatives

The detailed scorecard enables quick comparison across revenue, deals won, lost deals, and win rate.

Business Impact

Supports sales coaching, performance reviews, and incentive planning.

Dashboard 03

Product Intelligence Dashboard

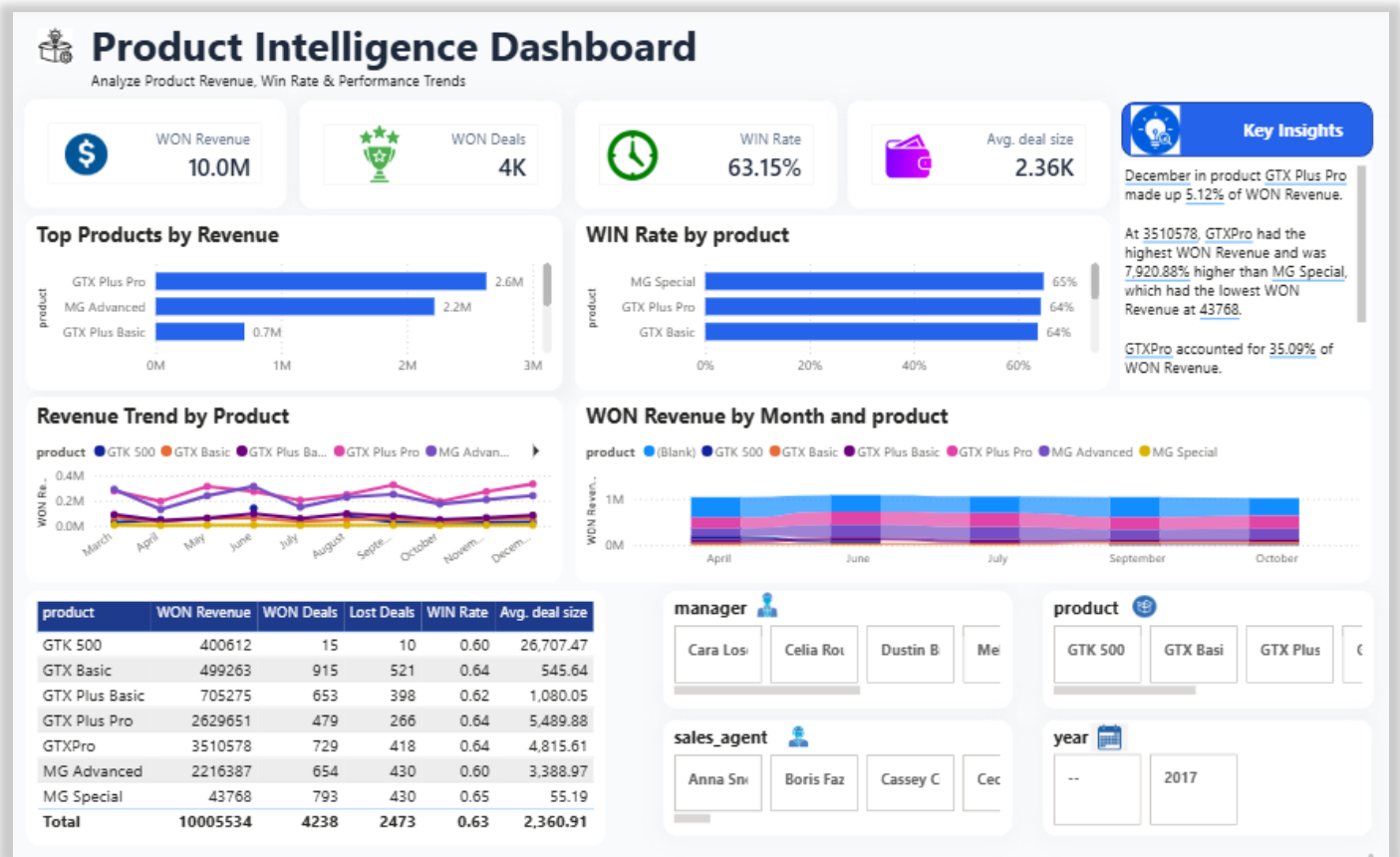


Figure 4: Product Intelligence Dashboard

Purpose

Understand product contribution towards organizational revenue.

Analysis

- Revenue by Product
- Win Rate by Product
- Monthly Product Revenue Trend
- Revenue Contribution
- Product Performance Table

AI-Based Business Recommendations

GTK Plus Pro contributes the highest revenue.

Revenue concentration suggests dependency on a limited number of products.

Lower-performing products require strategic review.

Management should prioritize investment in high-converting products while improving weaker product adoption.

Business Value

Supports product strategy and portfolio optimization.

Dashboard 04

Geographic Performance Dashboard

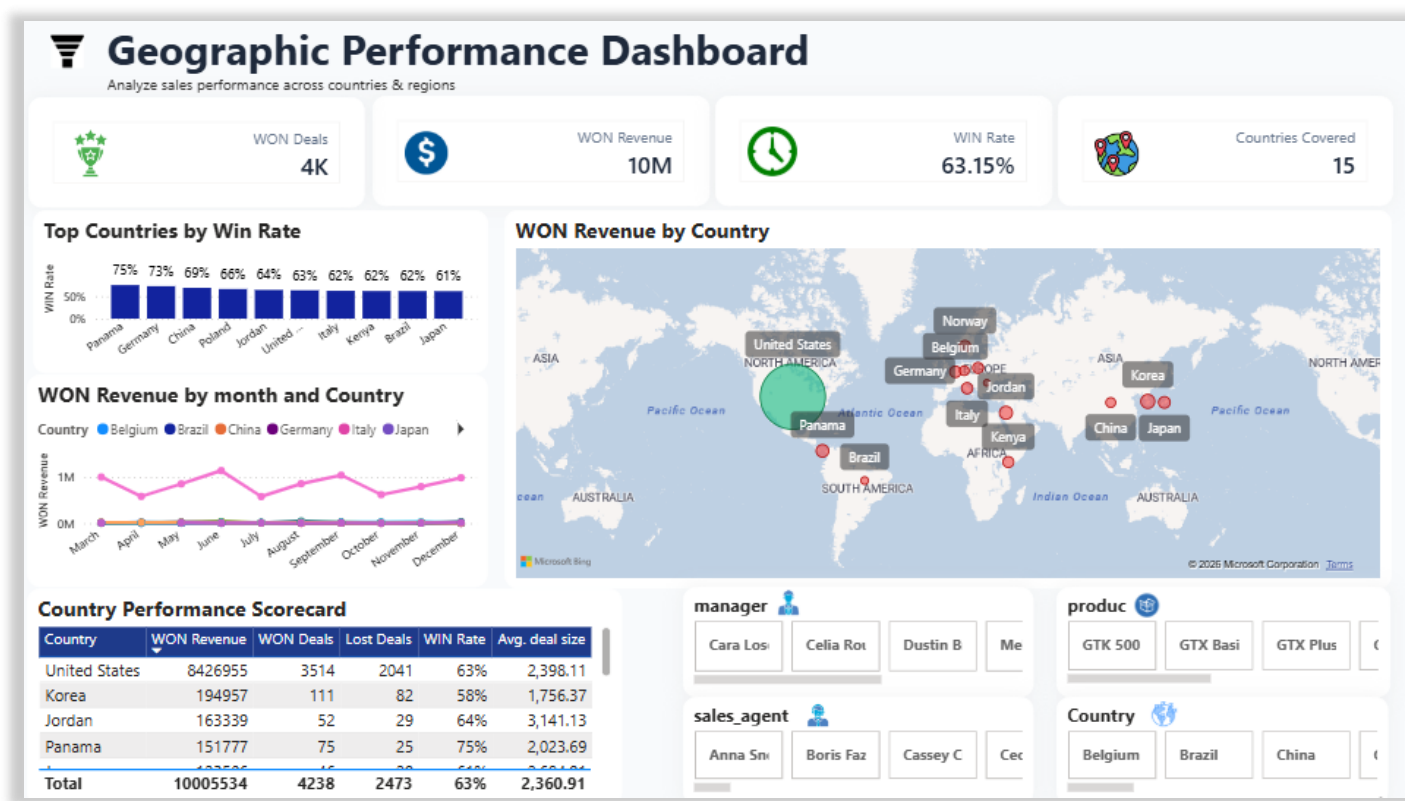


Figure 5: Geographic Performance Dashboard

Purpose

Analyze regional sales performance.

Visualizations

- Revenue Map
- Country Win Rate
- Monthly Country Revenue
- Country Performance Scorecard
- Interactive Geographic Filtering

Key Insights

The United States contributes the highest overall revenue.

Several countries exhibit high win rates despite relatively lower revenue, indicating growth opportunities.

The geographic map enables executives to identify regional strengths and expansion opportunities.

Business Value

Supports market expansion planning and regional investment decisions.

Dashboard 05

AI Insights & Recommendations

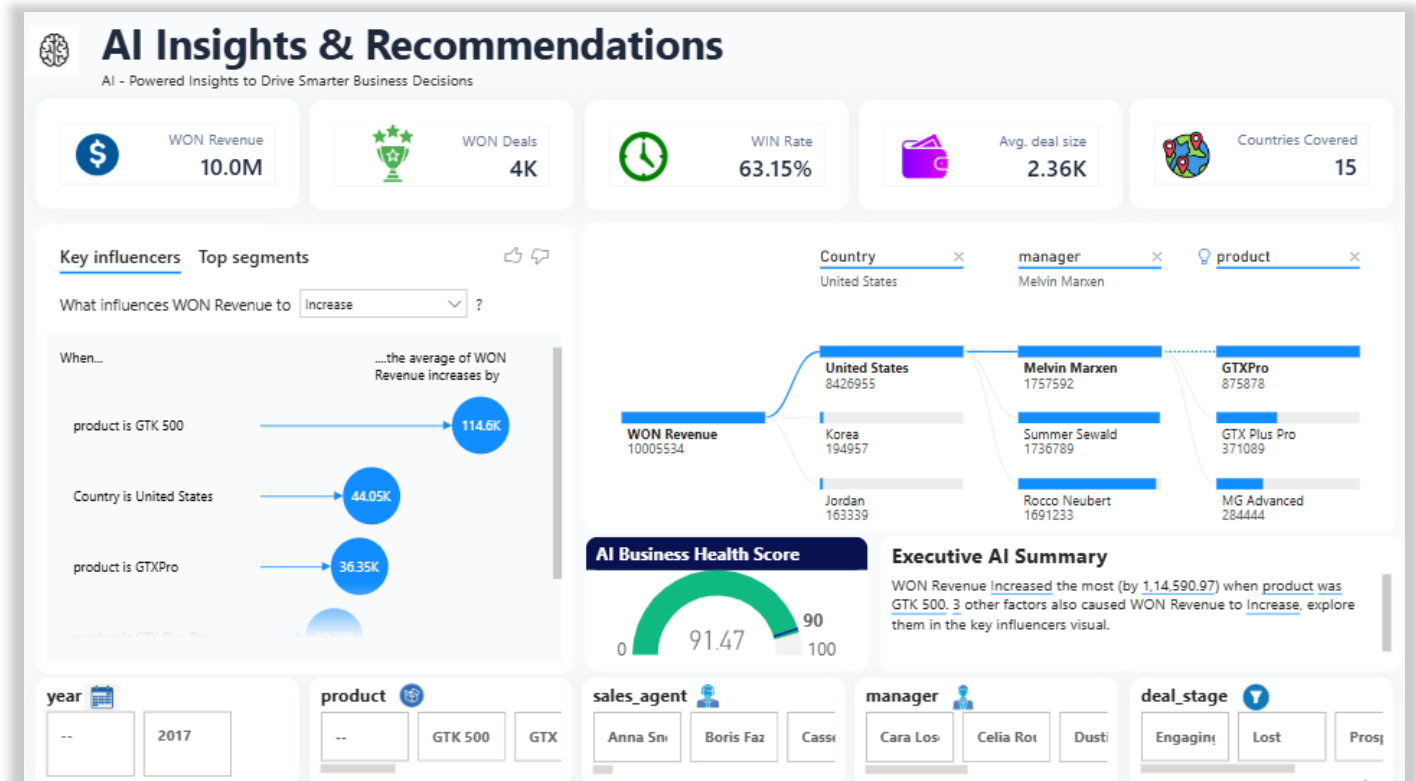


Figure 6: AI Insights & Recommendations Dashboard

AI Capabilities Implemented	
AI Feature	Business Purpose
Key Influencers	Identifies variables driving revenue changes
Decomposition Tree	Performs root-cause analysis
AI Summary	Generates executive insights
Business Health Score	Provides a quick performance indicator

AI Insights

Power BI AI visuals automatically identify variables that most significantly influence revenue performance.

The decomposition tree enables users to drill into country, manager, and product hierarchies to understand revenue contribution. The AI Business Health Score provides a summarized performance indicator, allowing executives to quickly assess business health.

Consultant Perspective

Rather than simply displaying charts, this dashboard explains the reasons behind business outcomes, supporting data-driven executive decision-making.

AI Value Delivered

Executives can identify revenue drivers and business opportunities within seconds instead of manually analyzing thousands of records.

Business Outcomes & Key Learnings

Business Outcomes

The project successfully transforms transactional CRM data into executive-level business intelligence.

The dashboard enables:

- Improved sales visibility
- Product performance analysis
- Country-wise market intelligence
- Sales representative evaluation
- AI-assisted business recommendations
- Interactive decision support

Technical Skills Demonstrated

Data Preparation

- Power Query
- Data Cleaning
- Data Transformation

Data Modeling

- Star Schema
- Relationships
- Date Table

Analytics

- DAX
- Time Intelligence
- KPI Development

Visualization

- Executive Dashboards
- Interactive Reports
- AI Visuals

Business

- Executive Reporting
- Business Storytelling
- Decision Support

Consultant Skills Demonstrated

- Business Requirement Gathering
- KPI Identification
- Executive Reporting
- Data Storytelling
- Analytical Thinking
- Strategic Recommendations
- Decision Support

Conclusion & Future Enhancements

Conclusion

This project demonstrates the practical application of Business Intelligence and AI-assisted analytics to transform CRM data into strategic insights. It reflects a combination of technical capability, business understanding, and data storytelling required to support executive decision-making.

Strategic Outcome: The dashboard enables leadership to monitor performance, identify growth opportunities, and support AI-assisted business decisions through a centralized analytics platform.

The project also reflects industry best practices in data modeling, visualization design, KPI development, and AI-powered analytics. And this project reflects my commitment to developing modern, AI-enabled Business Intelligence solutions that transform data into strategic business value.

Future Enhancements

- Predictive Sales Forecasting using Machine Learning
- Integration with Azure OpenAI
- Natural Language Query Support
- Copilot-Powered Dashboard Summaries
- Real-Time CRM Integration
- Power BI Service Deployment
- Row-Level Security (RLS)
- Automated Email Alerts
- Mobile Dashboard Optimization
- Executive AI Chat Assistant

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